Comprehensive Git Commands Cheat Sheet

git init

Options: [--bare]

Description: Initializes a new Git repository. Use --bare to create a repository without a working directory.

Example:

git init

or for a bare repo:

git init --bare

git clone

Options: [--depth <depth>] [--branch <branch>] [--single-branch]

Description: Clones a repository into a new directory. Supports shallow clones and specific branches.

Example:

git clone https://github.com/user/repo.git

git clone --depth 1 --branch main https://github.com/user/repo.git

git add

Options: [<pathspec>] [-A] [-u] [-p]

Description: Adds file contents to the staging area. Use -A to add all, -u to update tracked files, -p for interactive patching.

Example:

git add file.txt

git add -A

git add -p

git commit

Options: [-m <msg>] [--amend] [--no-edit]

Description: Records changes to the repository. Use -m to add a message, --amend to modify the last commit.

Example:

git commit -m 'Initial commit'

git commit --amend

git push

Options: [<remote>] [<branch>] [--force] [--tags]

Description: Updates remote refs along with associated objects. Use --force to overwrite, --tags to push tags.

Example:

git push origin main

git push --force

git push --tags

git pull

Options: [--rebase] [<remote>] [<branch>]

Description: Fetches from and integrates with another repository or branch. Use --rebase to reapply commits.

Example:

git pull origin main

git pull --rebase

git status

Options: [--short] [--branch]

Description: Displays paths that have differences between the index file and the current HEAD commit.

Example:

git status

git status --short

git log

Options: [--oneline] [--graph] [--decorate] [--stat]

Description: Shows the commit logs. Use --oneline for compact view, --graph for ASCII graph, --stat for file changes.

Example:

git log --oneline --graph --decorate

git branch

Options: [-a] [-d <branch>] [-m <old> <new>]

Description: Lists, creates, or deletes branches. Use -a to list all, -d to delete, -m to rename.

Example:

git branch

git branch -d feature

git branch -m old new

git checkout

Options: [<branch>] [<file>] [-b <new-branch>]

Description: Switches branches or restores working tree files. Use -b to create and switch to a new branch.

Example:

git checkout main

git checkout -b new-feature

git switch

Options: [<branch>] [-c <new-branch>]

Description: Switches branches. Use -c to create and switch to a new branch.

Example:

git switch main

git switch -c feature

git merge

Options: [<branch>] [--no-ff] [--squash]

Description: Joins two or more development histories together. Use --no-ff to always create a merge commit.

Example:

git merge feature-branch

git merge --no-ff feature

git rebase

Options: [<upstream>] [--interactive] [--onto]

Description: Reapplies commits on top of another base tip. Use --interactive for editing commits.

Example:

git rebase main

git rebase -i HEAD~3

git reset

Options: [--soft] [--mixed] [--hard] [<commit>]

Description: Resets current HEAD to the specified state. Use --soft to keep changes staged, --hard to discard all.

Example:

git reset --soft HEAD~1

git reset --hard

git revert

Options: [<commit>] [--no-edit]

Description: Creates a new commit that undoes the changes of a previous commit.

Example:

git revert abc123

git stash

Options: [save] [pop] [list] [apply]

Description: Temporarily shelves changes in the working directory.

Example:

git stash

git stash pop

git stash list

git tag

Options: [-a <tag>] [-d <tag>] [-l]

Description: Creates, lists, or deletes tags.

Example:

git tag -a v1.0 -m 'Version 1.0'

git tag -d v1.0

git fetch

Options: [<remote>] [--all] [--prune]

Description: Downloads objects and refs from another repository.

Example:

git fetch origin

git fetch --all

git remote

Options: [add <name> <url>] [remove <name>] [-v]

Description: Manages set of tracked repositories.

Example:

git remote -v

git remote add origin https://github.com/user/repo.git

git config

Options: [--global] [--list] [user.name|user.email]

Description: Gets and sets repository or global options.

Example:

git config --global user.name 'Alice'